Homework 5

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Problem 1

Let

$$B(\lambda)=\begin{pmatrix}\lambda^3+\lambda & \lambda^2\\ \lambda^2-3\lambda & 2\lambda\end{pmatrix},\qquad C(\lambda)=\begin{pmatrix}1-\lambda & \lambda^2 & \lambda\\ 2\lambda-1 & \lambda(1-\lambda) & -2\lambda\\ 1+\lambda^2 & \lambda^2 & -\lambda^2\end{pmatrix},$$

and

$$A(\lambda)=\begin{pmatrix}B(\lambda) & 0\\ 0 & C(\lambda)\end{pmatrix}.$$

Find out

(1) the Smith canonical form of $B(\lambda)$;

(2) the elementary divisors of $A(\lambda)$.

Solution

Work over $\mathbb{C}[\lambda]$ and let D_k denote the kth determinantal divisor.

(1) Every entry of B is divisible by λ, and 2λ is an entry, so $D_1(B)=\lambda$. Also

$$\det B=\lambda^2(\lambda+1)(\lambda+2),\qquad \boxed{S_B=\operatorname{diag}\big(\lambda,\lambda(\lambda+1)(\lambda+2)\big)}.$$

(2) For C, direct calculation gives $D_1=1$, $D_2=\lambda$ and $D_3=\lambda^2(\lambda+1)$, hence

$$S_C=\operatorname{diag}\big(1,\lambda,\lambda(\lambda+1)\big).$$

Take the multiset union of the elementary divisors of both blocks. All elementary divisors of A are

$$\boxed{\lambda,\ \lambda,\ \lambda,\ \lambda,\ \lambda+1,\ \lambda+1,\ \lambda+2}.$$

Problem 2

Let $D:\mathbb{C}[\lambda]\to\mathbb{C}[\lambda]$ be defined by

$$D\left(\sum_{i=0}^{n}\alpha_i\lambda^i\right)=\sum_{i=1}^{n}i\alpha_i\lambda^{i-1}.$$

Show that D is a linear transformation on $\mathbb{C}[\lambda]$, and determine $\ker[D]$.

Solution

Differentiation maps polynomials to polynomials and satisfies $D(uf+vg)=uD(f)+vD(g)$ for polynomials f,g and complex scalars u,v, so D is linear. Moreover, $D(f)=0$ iff $i\alpha_i=0$ for every $i\geq 1$, which means f is constant. Thus

$$\boxed{\ker D=\mathbb{C}\cdot 1=\operatorname{span}\{1\}}.$$

Problem 3

Let $\mathcal{A} \in L(\mathbb{C}_n[\lambda])$ be defined, for a nonzero constant $a \in \mathbb{C}$, as

$$\mathcal{A}[f(\lambda)] = f(\lambda + a), \qquad \forall f(\lambda) \in \mathbb{C}_n[\lambda].$$

Find out

(1) the matrix representation of $\mathcal{A}$ with respect to the basis $1, \lambda, \lambda^2, \ldots, \lambda^n$;

(2) the eigenvalues and the associated eigenvectors of $\mathcal{A}$.

Solution

(1) By $(\lambda + a)^j = \sum_{i=0}^{j} \binom{j}{i} a^{j-i} \lambda^i$, the matrix $M = [\mathcal{A}]$, with rows and columns indexed from 0 to n, is

$$\boxed{M_{ij} = \begin{cases} \binom{j}{i} a^{j-i}, & 0 \le i \le j \le n, \\ 0, & i > j. \end{cases}}$$

(2) M is upper triangular with all diagonal entries one, so its only eigenvalue is 1, of algebraic multiplicity $n+1$. If f has degree $m \ge 1$ and leading coefficient c_m, then $f(\lambda + a) - f(\lambda)$ has nonzero leading term $m c_m a \lambda^{m-1}$. Thus only constant polynomials satisfy $f(\lambda + a) = f(\lambda)$, and

$$\boxed{E_1(\mathcal{A}) = \operatorname{span}\{1\}, \qquad f(\lambda) = c \ (c \in \mathbb{C} \setminus \{0\})}.$$

The nonzero constant polynomials are precisely all associated eigenvectors.

Problem 4

Let $V = \mathbb{C}_n[s]$ and $D : \mathbb{C}_n[s] \to \mathbb{C}_n[s]$ be defined by

$$D[f(s)] = n a_n s^{n-1} + (n-1) a_{n-1} s^{n-2} + \cdots + 2 a_2 s + a_1,$$
$$f(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_2 s^2 + a_1 s + a_0.$$

Find out

(1) the elementary divisors of D;

(2) the eigenvalues and the associated eigenvectors of D.

Solution

(1) In the basis $(1, s, \ldots, s^n)$, $N = [D]$ has nonzero entries only at $N_{j-1,j} = j$ for $1 \le j \le n$. Use μ as the characteristic-matrix indeterminate. The superdiagonal of $\mu I - N$ is $-1, -2, \ldots, -n$. For each $1 \le k \le n$ there is a nonzero constant minor $(-1)^k k!$, so $D_k = 1$; also $\det(\mu I - N) = \mu^{n+1}$. Hence

$$S_{\mu I - N} = \operatorname{diag}\big(\underbrace{1, \ldots, 1}_{n}, \mu^{n+1}\big), \qquad \boxed{\text{the sole elementary divisor is } \mu^{n+1}}.$$

(2) N is strictly upper triangular, so its only eigenvalue is 0, of algebraic multiplicity $n+1$. Since $D(f) = f' = 0$ exactly for constant polynomials,

$$\boxed{E_0(D) = \operatorname{span}\{1\}, \qquad f(s) = c \ (c \in \mathbb{C} \setminus \{0\})}.$$

All associated eigenvectors are the nonzero constant polynomials.